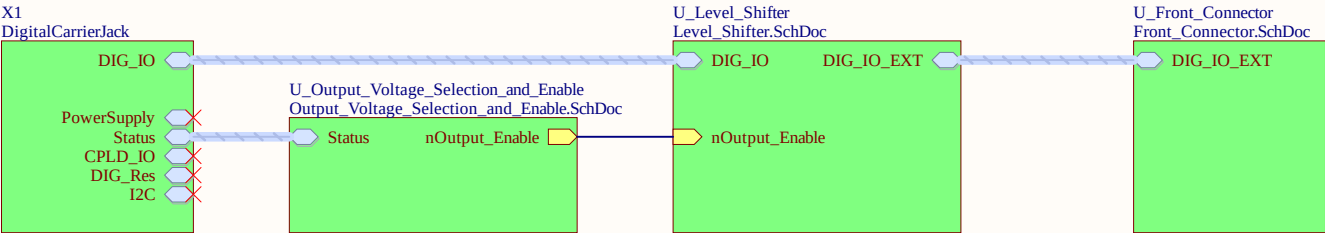
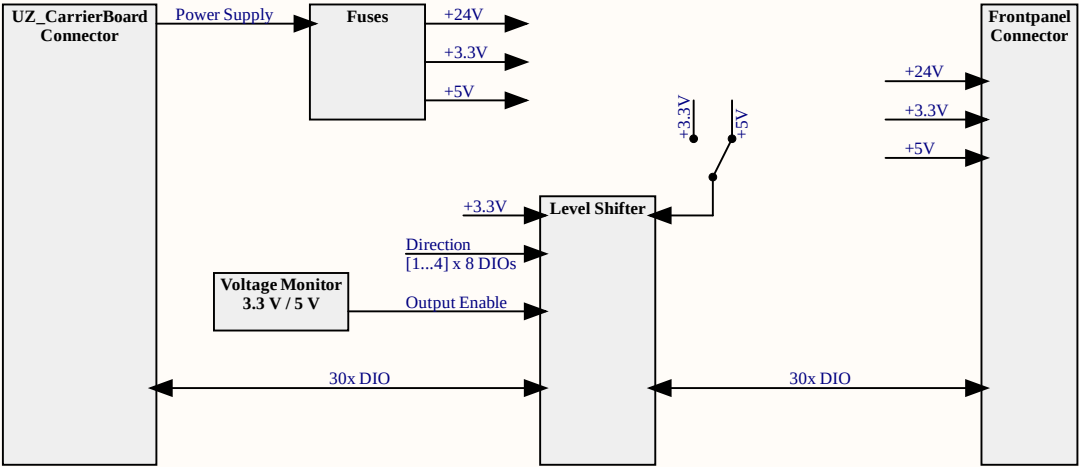


Top Sheet



Block Diagram



Chose size of UZ logo
by chosing footprint
type

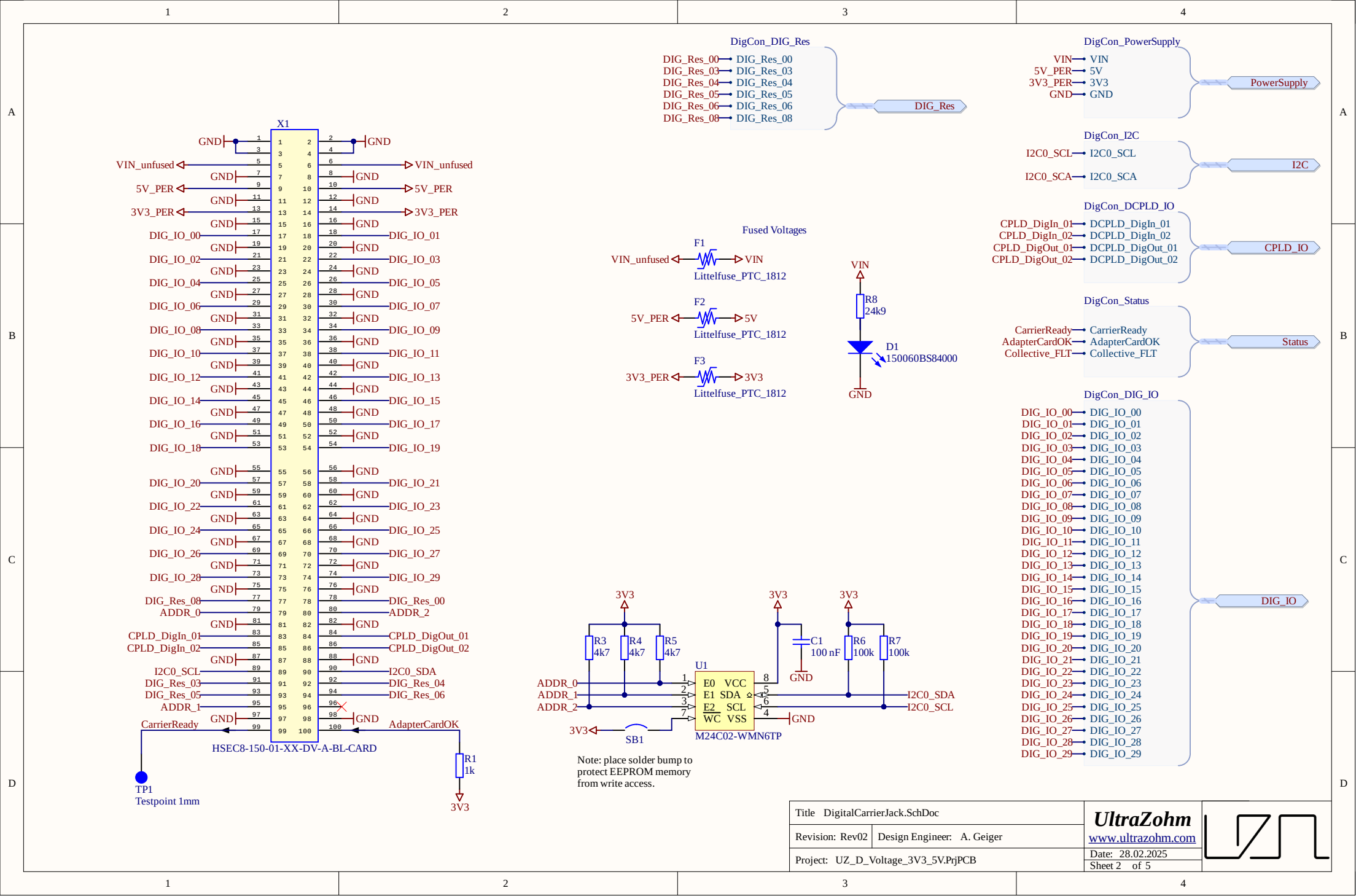


INFO

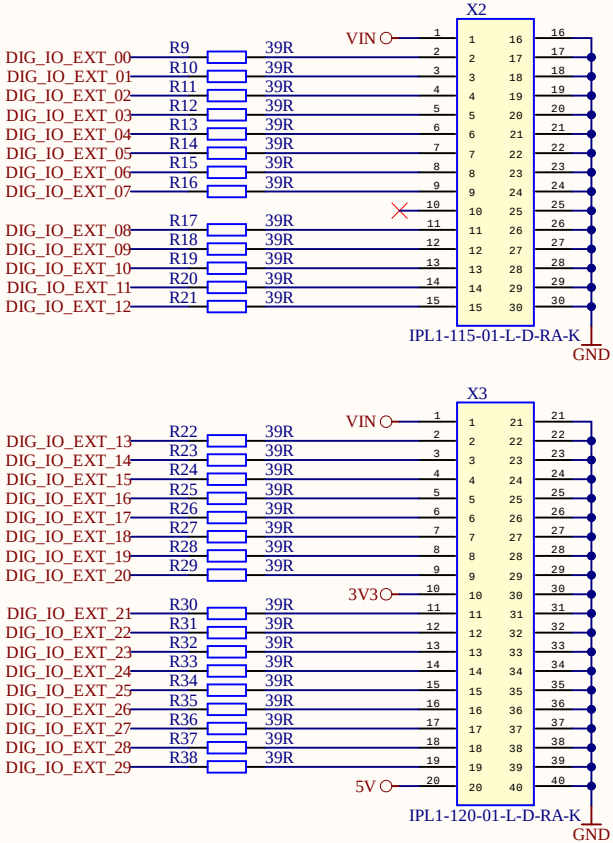
Project
ProjectRevision
AuthorParam
ProjectDate

Design Information

Title TopSheet.SchDoc		UltraZohm www.ultrazohm.com
Revision: Rev02	Design Engineer: A. Geiger	
Project: UZ_D_Voltage_3V3_5V.PrjPCB		
		Date: 28.02.2025
		Sheet 1 of 5



Front Panel Connector

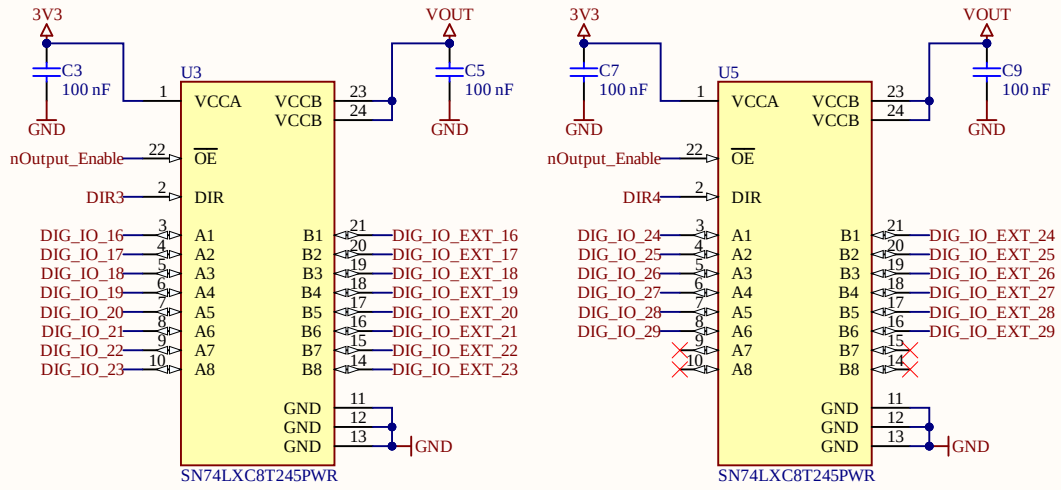
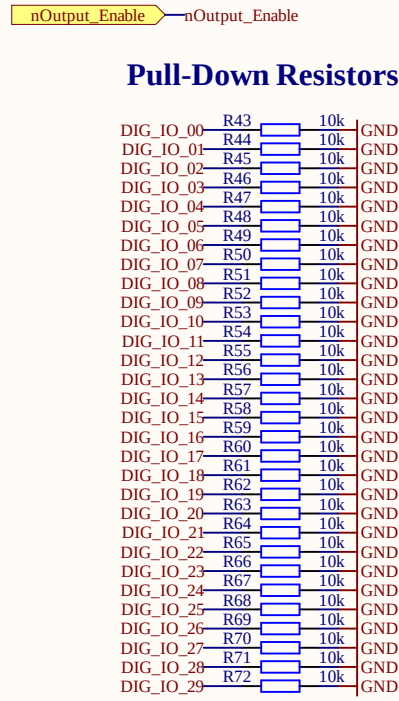
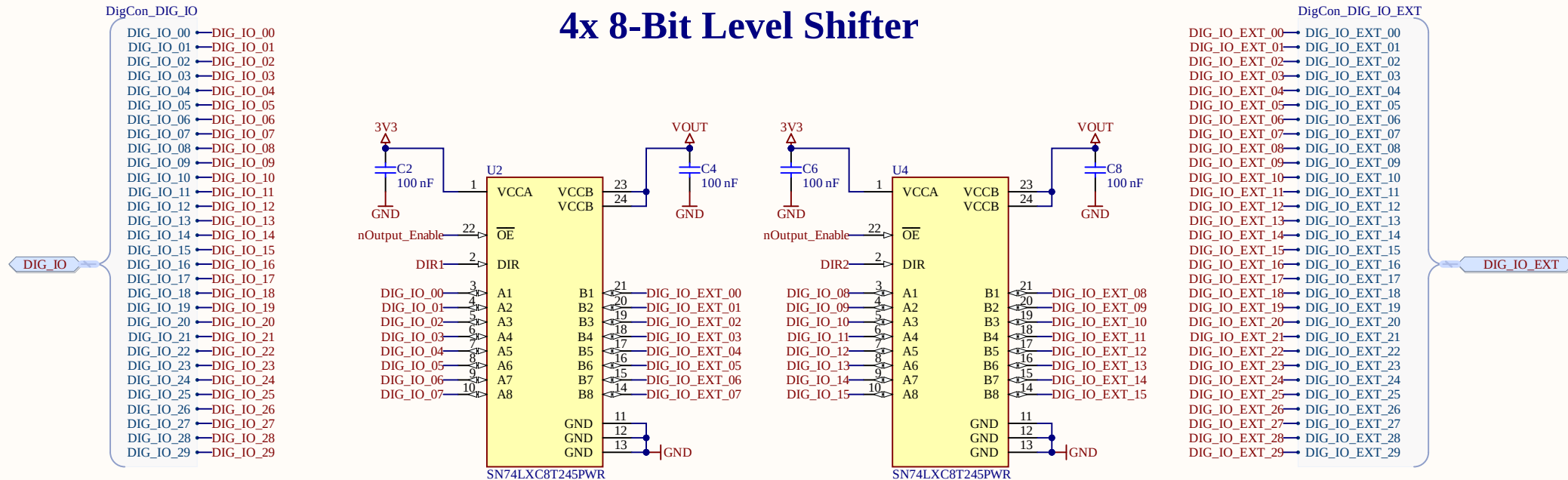


DigCon_DIG_IO_EXT

- DIG_IO_EXT_00
- DIG_IO_EXT_01
- DIG_IO_EXT_02
- DIG_IO_EXT_03
- DIG_IO_EXT_04
- DIG_IO_EXT_05
- DIG_IO_EXT_06
- DIG_IO_EXT_07
- DIG_IO_EXT_08
- DIG_IO_EXT_09
- DIG_IO_EXT_10
- DIG_IO_EXT_11
- DIG_IO_EXT_12
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- DIG_IO_EXT_23
- DIG_IO_EXT_24
- DIG_IO_EXT_25
- DIG_IO_EXT_26
- DIG_IO_EXT_27
- DIG_IO_EXT_28
- DIG_IO_EXT_29

DIG_IO_EXT

4x 8-Bit Level Shifter



Direction per Level Shifter

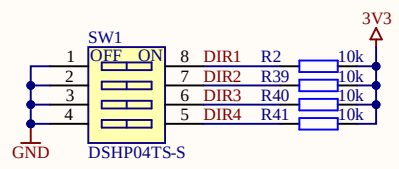


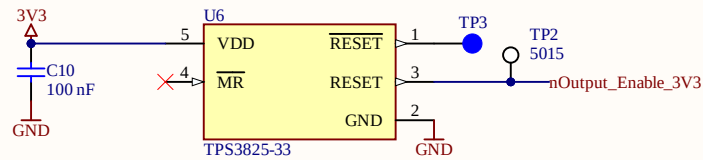
Table 8-1. Function Table⁽¹⁾

CONTROL INPUTS		Port Status		OPERATION
OE	DIR	A PORT	B PORT	
L	L	Output (Enabled)	Input (Hi-Z)	B data to A bus
L	H	Input (Hi-Z)	Output (Enabled)	A data to B bus
H	X	Input (Hi-Z)	Input (Hi-Z)	Isolation

(1) Input circuits of the data I/Os are always active and should be kept at a valid logic level.

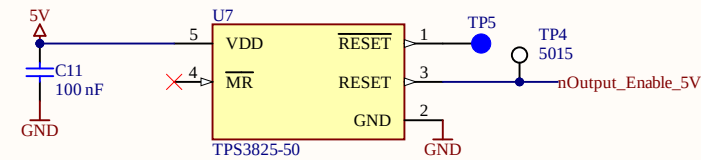
DIP-Switch	DIR	Signal Direction
OFF	HIGH	Output
ON	LOW	Input

3.3 V Voltage Monitoring



3V3 Rail	nOutput_Enable_3V3	Level Shifter Outputs
0 ... 1,1 V	LOW	enabled
1,1 ... 2,93 V	HIGH	disabled
2,93 ... 3,3 V	LOW	enabled

5 V Voltage Monitoring



5V Rail	nOutput_Enable_5V	Level Shifter Output
0 ... 1,1 V	LOW	enabled
1,1 ... 4,55 V	HIGH	disabled
4,55 ... 5 V	LOW	enabled

Voltage Monitoring Functionality

Table 6-1. Pin Functions

PIN				I/O	DESCRIPTION
NAME	TPS3820, TPS3823, TPS3823A, TPS3828	TPS3824	TPS3825		
GND	2	2	2	—	Ground connection
MR	3	—	4	I	Manual-reset input. Pull low to force a reset. RESET remains low as long as MR is low and for the time-out period after MR goes high. Leave unconnected or connect to V _{DD} when unused.
RESET	—	3	3	O	Active-high reset output. Either push-pull or open-drain output stage.
RESET	1	1	1	O	Active-low reset output. Either push-pull or open-drain output stage.
V _{DD}	5	5	5	I	Supply voltage. Powers the device and monitors its own voltage.
WDI	4	4	—	I	Watchdog timer input. If WDI remains high or low longer than the time-out period, then reset is triggered. The timer clears when reset is asserted or when WDI sees a rising edge or a falling edge. If unused, the WDI connection must be high impedance to prevent it from causing a reset event.

8.4 Device Functional Modes

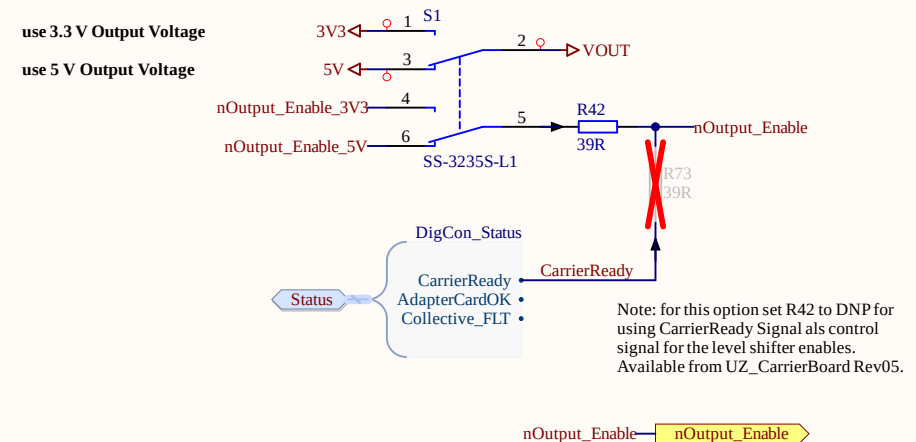
Table 8-1 lists the functional modes of the TPS382x devices.

Table 8-1. Function Table

INPUTS		OUTPUTS	
MR ⁽¹⁾	$V_{DD} > V_{IT}$	RESET	RESET ⁽²⁾
L	0	L	H
L	1	L	H
H	0	L	H
H	1	H	L

(1) TPS3820/3/5/8
(2) TPS3824/5

Level Shifter - Output Voltage Selection



Title Output_Voltage_Selection_and_Enable.SchDoc		 
Revision: Rev02	Design Engineer: A. Geiger	
Project: UZ_D_Voltage_3V3_5V.PrjPCB		
		Date: 28.02.2025 Sheet 5 of 5